

Titanic Survival Prediction

Using Machine Learning to Classify Passenger Survival

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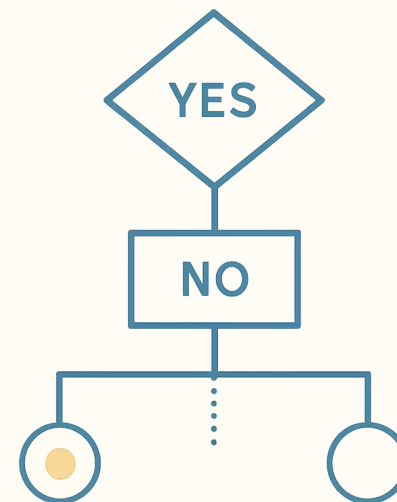
Group 14

PROBLEM STATEMENT

Predicting Survival on the Titanic

The Titanic dataset presents a classic **binary classification** problem: given passenger attributes, can we accurately predict who survived?

- **ML Type:** Supervised Classification
- **Target Variable:** Survived — 0 = Did Not Survive, 1 = Survived
- **Goal:** Train a model to generalize survival patterns from historical data



Kaggle Titanic Dataset

The dataset was sourced from the Kaggle Titanic competition — a benchmark challenge for classification modeling.

Training Data

891 rows × 12 columns

Includes the target variable Survived. Used to fit and validate models.

Test Data

418 rows × 11 columns

Missing the target column. Used for final predictions and Kaggle submission.

Key Features Selected

Six passenger attributes were identified as the most predictive for survival classification.



Pclass

Passenger ticket class (1st, 2nd, 3rd)



Sex

Male or female



Age

Passenger age in years



Fare

Ticket price paid



Embarked

Port of embarkation (C, Q, S)



FamilySize

SibSp + Parch + 1

Handling Missing Values

Missing Values Found

177

Age

Missing entries

687

Cabin

Missing entries

2

Embarked

Missing entries

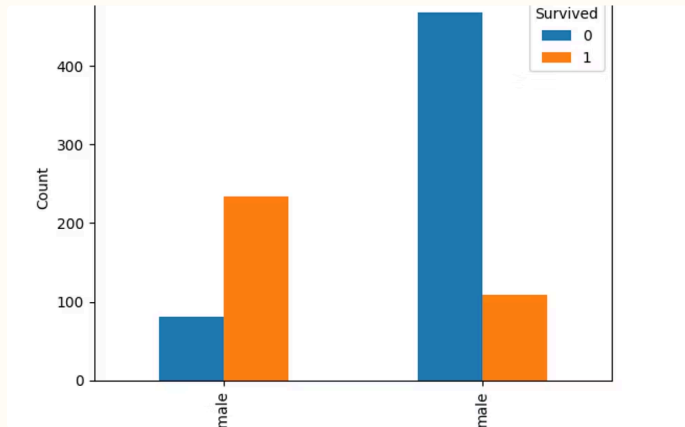
Resolution Strategy

→ **Age → Fill with Mean**
Replace 177 missing ages with the column mean to preserve row count.

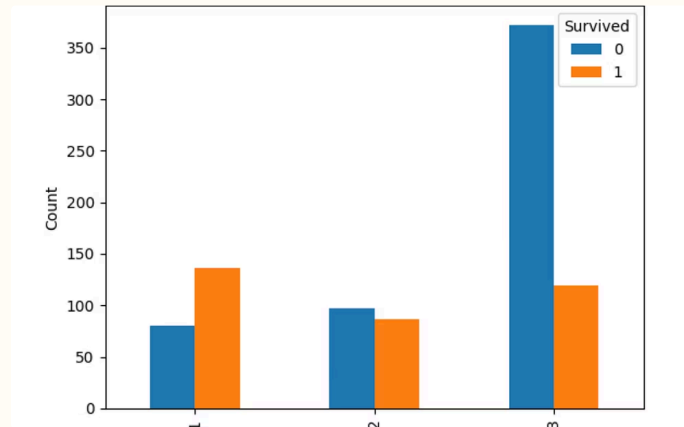
→ **Embarked → Fill with Mode**
Only 2 missing — replaced with the most frequent port value.

→ **Cabin → Drop Column**
Over 77% missing — too sparse to impute reliably; removed entirely.

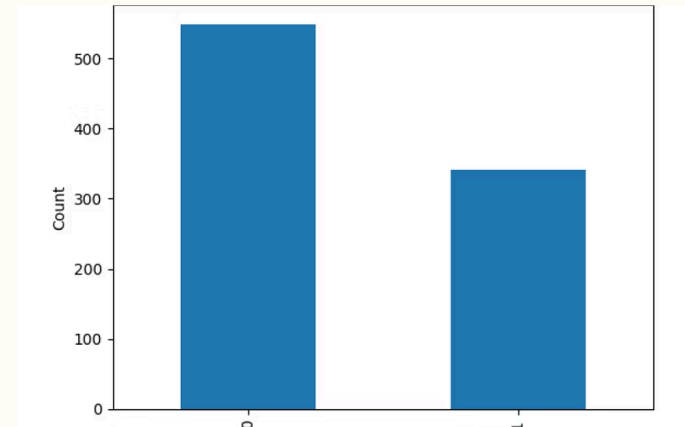
Key Survival Patterns



Females survived more than males

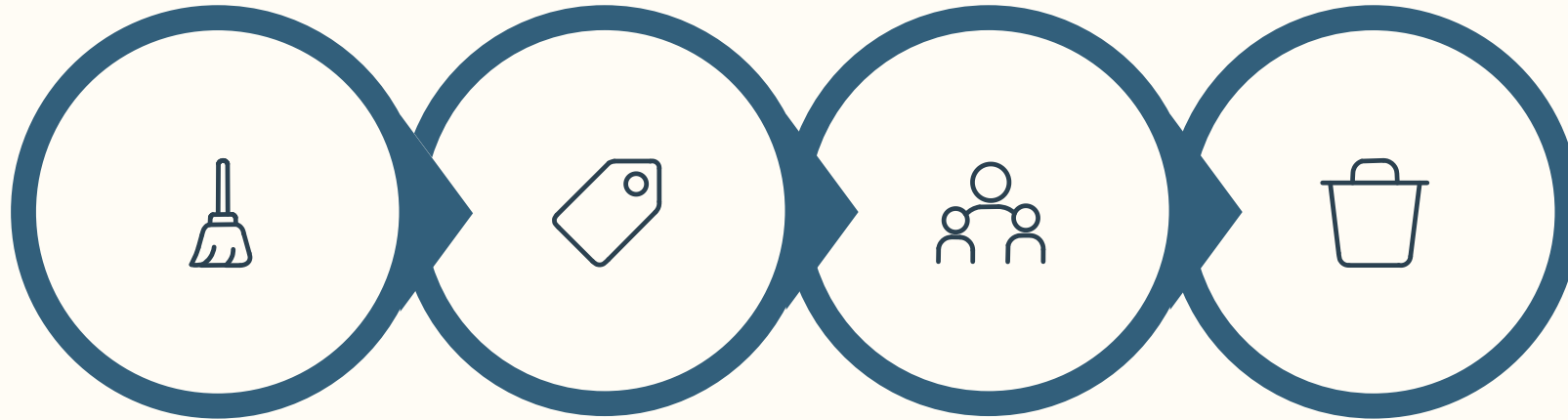


First-class passengers survived more



Higher fare passengers had better survival chances

Data Preprocessing



Data Preprocessing
- Data Preprocessing

Data Preprocessing
- Data Preprocessing

Create familySize
- combined Sibsp
and parch

Drop culomn
-Removed Cabin

Preprocessing steps

Data Preprocessing

Data Preprocessing

Data Preprocessing

Data Preprocessing

Models Used

Three supervised learning algorithms were trained and compared on the preprocessed Titanic dataset.

1. Logistic Regression

- Linear classification model
- Simple and interpretable

2. Decision Tree

- Tree-based model
- Easy to visualize

3. Random Forest

- Ensemble of decision trees
- Stronger performance

Hyperparameter Tuning with GridSearchCV

GridSearchCV

Systematically tests combinations of hyperparameters using cross-validation to find the best-performing model settings.

Best Parameters Found

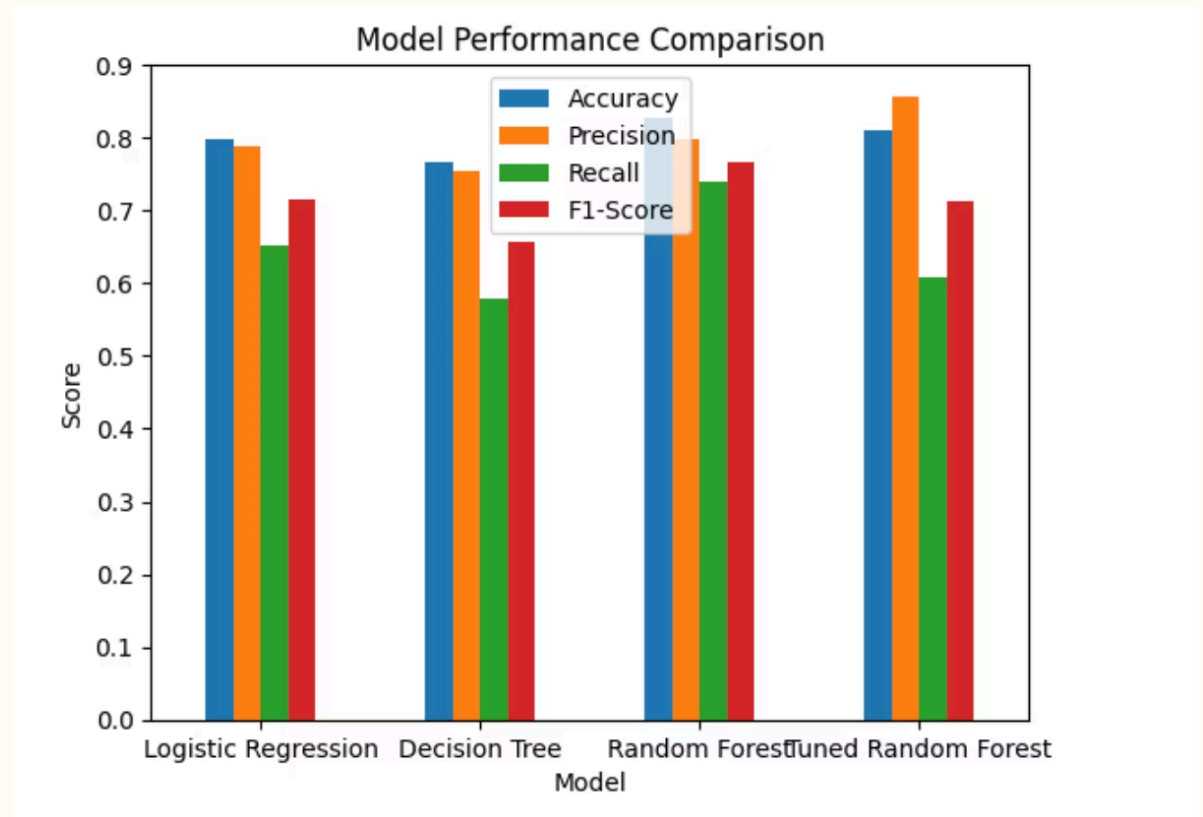
`max_depth=4`

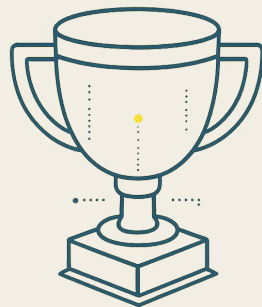
`n_estimators=10`

Model Comparison

Understanding the key metrics used to evaluate classification model performance.

- Precision
- Accuracy
- F1-score
- Recall





Best Model

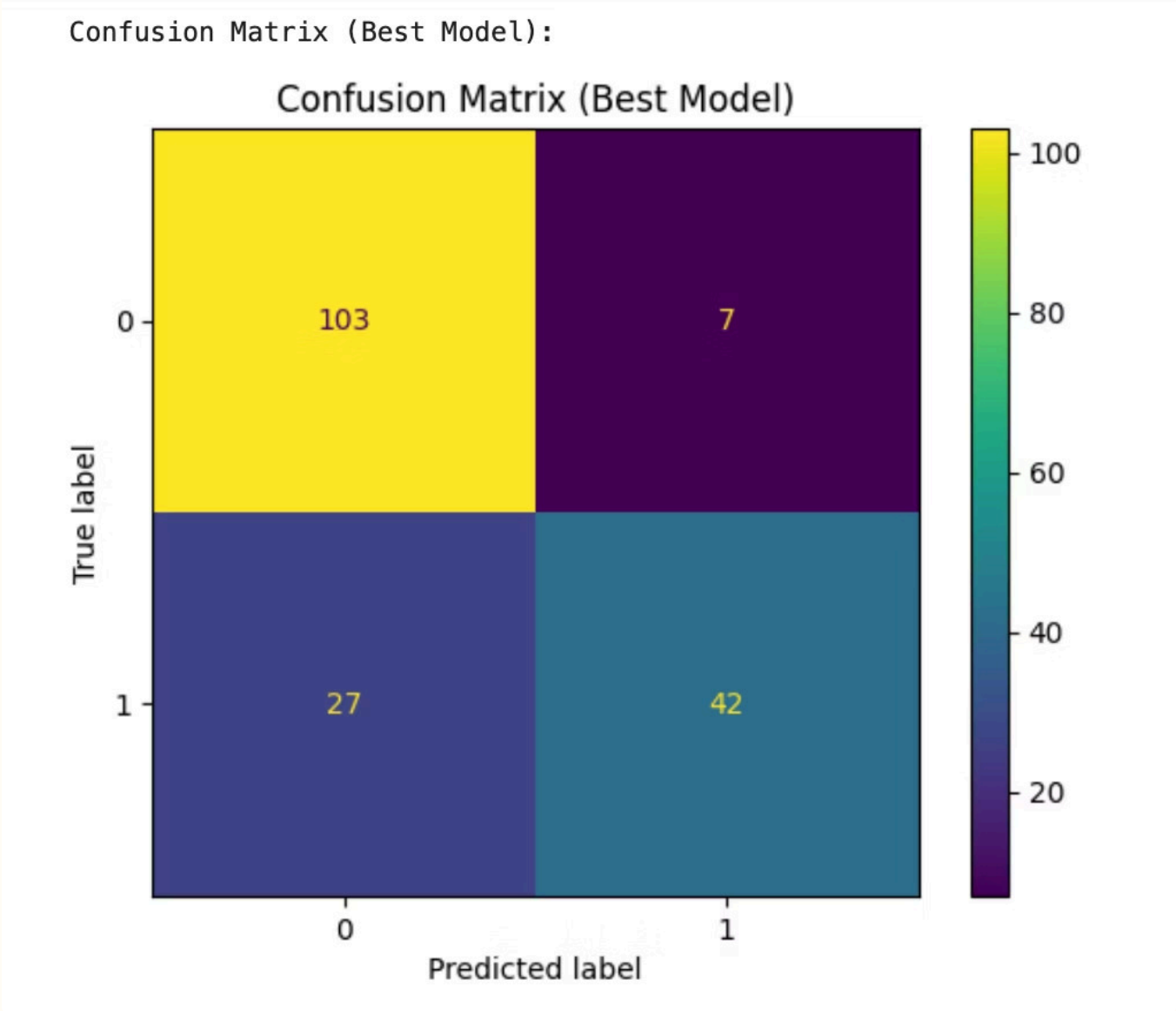
Tuned Random Forest

Why?

- Highest accuracy
- Better balanced performance
- Improved using GridSearchCV

Confusion Matrix

Confusion Matrix of Best Model



CONCLUSION

Conclusion

- Titanic survival prediction is a classification problem
- Data preprocessing improved the model
- Tuned Random Forest performed best
- Machine learning can effectively predict passenger survival

THANK YOU

Thank You 🎉

Questions? We're All Ears! 👂

Thanks for sailing with us — happy to answer anything!